

3.5 (b).

$$x^\mu(\lambda) = (c, \lambda).$$

$$\Rightarrow \frac{dx^\sigma}{d\lambda} = (0, 1)$$

$$\frac{D}{d\lambda} V = 0$$

$$\Rightarrow \frac{dx^\sigma}{d\lambda} \nabla_\sigma V^\rho = 0.$$

$$\Rightarrow \nabla_\phi V^\rho = 0.$$

$$\Rightarrow \partial_\phi V^\rho + \Gamma_{\phi\alpha}^\rho V^\alpha = 0.$$

$$\Rightarrow \partial_\phi V^\rho + \Gamma_{\phi 0}^\rho V^0 + \Gamma_{\phi 1}^\rho V^1 = 0.$$

2 eqn-s:

$$\textcircled{1} \partial_\phi V^0 + \Gamma_{\phi 0}^0 V^0 + \Gamma_{\phi 1}^0 V^1 = 0.$$

$$\textcircled{2} \partial_\phi V^1 + \Gamma_{\phi 0}^1 V^0 + \Gamma_{\phi 1}^1 V^1 = 0.$$

$$\textcircled{1} \quad \partial_\phi V^0 - \sin\theta \cos\theta V^1 = 0.$$

$$\textcircled{2} \quad \partial_\phi V^1 + \cot\theta V^0 = 0.$$

$$\Gamma_{\nu\rho}^0 = \begin{pmatrix} 0 & 0 \\ 0 & -\sin\theta \cos\theta \end{pmatrix}$$

$$\Gamma_{\nu\rho}^1 = \begin{pmatrix} 0 & \cot\theta \\ \cot\theta & 0 \end{pmatrix}$$

If $\theta = \pi/2$, $V^\mu = (C_1, C_2)$ where C_1 & C_2 are independent of ϕ .

So $V^\mu = (1, 0)$, using initial conditions.

$$\text{Else, } V^1 = \frac{\partial_\phi V^0}{\sin\theta \cos\theta}$$

$$\Rightarrow \frac{1}{\sin\theta \cos\theta} \partial_\phi^2 V^0 + \cot\theta V^0 = 0.$$

$$\Rightarrow \partial_\phi^2 V^0 = -\cos^2\theta V^0$$

$$\Rightarrow V^0 = A \cos(\phi \cos \theta + \delta)$$

$$\partial_\theta V^0 = -A \sin(\phi \cos \theta + \delta) \cos \theta$$

$$\Rightarrow \partial_\phi V^0 - \sin \theta \cos \theta V^1$$

$$= -A \sin(\phi \cos \theta + \delta) \cancel{\cos \theta} - \sin \theta \cancel{\cos \theta} V^1 = 0.$$

$$\Rightarrow V^1 = -A \frac{\sin(\phi \cos \theta + \delta)}{\sin \theta}$$

When $\lambda=0$, $V^\mu = (1, 0)$, & $\phi=0$.

$$\text{So } V^0(\lambda=0) = A \cos \delta = 1$$

$$V^1(\lambda=0) = -A \frac{\sin \delta}{\sin \theta} = 0.$$

$$\text{So, } \sin \delta = 0 \Rightarrow \delta = 0 \text{ or } \pi \Rightarrow A = 1 \text{ or } -1.$$

Both routes give the same thing. So:

$$\begin{pmatrix} V^0 \\ V^1 \end{pmatrix} = \begin{pmatrix} \cos(\phi \cos \theta) \\ -\sin(\phi \cos \theta) \csc \theta \end{pmatrix}$$

we can set $\phi = \lambda$.